Basin parameters

SWAT+ Documentation

docs.swat.tamu.edu

Contents

Purpose . . . 2
Format . . . 2
PET method . . . 4
Extra input columns . . . 4
Crack flow . . . 4
Peak runoff rate . . . 4
Channel routing . . . 5
USLE C factor . . . 5
Carbon model . . . 5
Lapse rates . . . 5
Unit hydrograph . . . 6
Runoff method . . . 6
In-stream nutrients . . . 6
gwflow module . . . 6
Tillage effect on decomposition . . . 6
Example . . . 7
Related files . . . 7
Purpose . . . 8
Format . . . 8
Zero and missing values . . . 12
Initial soil water . . . 12
Surface runoff lag . . . 12
Peak rate factor . . . 12
Pesticides in aquifers . . . 13
Pesticide decay and temperature . . . 13
Uptake distribution . . . 13
Denitrification . . . 13
Frozen soil . . . 14
Dormancy threshold . . . 14
USLE C factor . . . 14
Sub-daily erosion . . . 14
CO2 . . . 14
Random numbers . . . 14
Calibration . . . 15
Example . . . 15
Related files . . . 15
Purpose . . . 17
Activation and file names . . . 17
Format . . . 17
Initial carbon pools . . . 19
Soluble carbon losses . . . 19
Organic fertilizer and manure . . . 19
Tillage and biological mixing . . . 19
Temperature factor . . . 20

Water factor . . . 20
Residue decomposition . . . 20
Notes on individual columns . . . 21
Calibration . . . 21
Example . . . 21
Related files . . . 21
Purpose . . . 22
Activation and file name . . . 22
Format . . . 22
Use in the carbon model . . . 24
Calibration . . . 24
Example . . . 24
Related files . . . 25

Purpose

`codes.bsn` contains the method codes that apply to the whole watershed: the potential evapotranspiration (PET) method, the channel routing method, the soil carbon model, the surface runoff method, the tile drainage equations, plant stress switches and a few output options. It is named in the first position of the `basin` line of `file.cio` (default `codes.bsn`). SWAT+ reads it at the start of the run, before the inputs that depend on these codes.

If the file named in `file.cio` does not exist, every code keeps the built-in value listed below.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Line 3 contains 26 values in the order below. The reader matches values by position only, so all 26 must be present and in this order. The table gives the column headers the SWAT+ Editor writes.

#	Column	Type	Built-in value	Description
1	`pet_file`	text	`pet.cli`	Not used. Measured PET is set up through `file.cio` and `weather-sta.cli` (see PET method).
2	`wq_file`	text	blank	Not used.
3	`pet`	integer	0	PET method: `0` Priestley-Taylor, `1` Penman-Monteith, `2` Hargreaves, `3` PET of the weather station. See PET method.
4	`event`	integer	0	Extra column switch for two other inputs. `0`: normal format. Any other value: `plants.plt` is read with an extra plant class column and `soil_plant.ini` with an extra constituent column. See Extra input columns.
5	`crack`	integer	0	Crack flow: `0` off, `1` on. See Crack flow.
6	`swift_out`	integer	0	SWIFT input files: `0` not written, `1` written to a `SWIFT` folder at the end of the run. Older Editor versions label this column `rtu_wq`.
7	`sed_det`	integer	0	Peak runoff rate method for erosion: `0` NRCS method with the peak rate factor, `1` half-hour rainfall intensity method. See Peak runoff rate.
8	`rte_cha`	integer	0	Channel water routing: `0` variable storage, `1` Muskingum. See Channel routing.
9	`deg_cha`	integer	0	Not used.

#	Column	Type	Built-in value	Description
10	`wq_cha`	integer	0	Not used.
11	`nostress`	integer	0	Plant stress: `0` all stresses applied, `1` all stresses turned off (water, temperature, nitrogen, phosphorus, aeration and salt), `2` nitrogen and phosphorus stress turned off.
12	`cn`	integer	0	Not used.
13	`c_fact`	integer	0	Read, then overridden: the model always uses the residue and biomass cover method for the USLE C factor. See USLE C factor.
14	`carbon`	integer	0	Soil carbon model: `0` static carbon (original SWAT humus and residue routines), `1` reserved for C-FARM (not implemented), `2` dynamic carbon model with CENTURY pools. See Carbon model.
15	`lapse`	integer	0	Elevation adjustment of HRU precipitation and temperature: `0` off, `1` on. See Lapse rates.
16	`uhyd`	integer	1	Unit hydrograph shape for sub-daily runs: `0` triangular, `1` gamma function. See Unit hydrograph.
17	`sed_cha`	integer	0	Not used.
18	`tiledrain`	integer	0	Tile flow equation: `0` drawdown-days equation, `1` DRAINMOD equations.
19	`wtable`	integer	0	Shallow water table depth: `0` original routine (water above field capacity fills the profile from the bottom), `1` DRAINMOD water table routine.
20	`soil_p`	integer	0	Soil phosphorus model: `0` original SWAT model with a fixed P availability index, `1` dynamic P availability of Vadas and White (2010).
21	`gampt`	integer	0	Infiltration and surface runoff: `0` curve number, `1` Green-Ampt. See Runoff method.
22	`atmo_dep`	single character	`a`	Not used.
23	`stor_max`	integer	0	Not used.
24	`i_fpwet`	integer	0	In-stream nutrient method: `0` QUAL2E, `1` QUAL2E followed by simplified first-order transformations. Called `qual2e` in earlier documentation. See In-stream nutrients.
25	`gw_flow`	integer	0	Gridded groundwater module: `0` off, `1` on. Called `gwflow` in earlier documentation, where it was described as flood routing. See gwflow module.
26	`idc_till`	integer	3	Tillage effect on decomposition in the dynamic carbon model (`carbon = 2`): `1` DSSAT, `2` EPIC (placeholder), `3` Kemanian, `4` DNDC (placeholder). See Tillage effect on decomposition.

The 26 values are read in one free-format read. A line with fewer values continues onto the next line of the file, and the codes that are not reached keep their built-in values. A value that cannot be read (text in an integer column, for example) ends the read without a message, so the codes after it also keep their built-in values.

PET method

`pet` selects how daily potential evapotranspiration is calculated for each HRU:

`pet`	Method
0	Priestley-Taylor, with $\alpha = 1.28$.
1	Penman-Monteith, with a 40 cm alfalfa reference crop and a canopy resistance that depends on the year's CO2 concentration. This method also gives the maximum transpiration of the growing plants.
2	Hargreaves.
3	The daily PET value of the HRU's weather station.

In every case the result is multiplied by the HRU's `pet_co` from `hydrology.hyd`.

With `pet = 3`, the HRU uses the daily PET stored for its weather station. A station gets measured PET from the PET gage named in `weather-sta.cli`, and the gage files are listed in `pet.cli`, which is named on the `climate` line of `file.cio` (see climate files). In 62.0.1, however, the station PET is recalculated with the Hargreaves equation later in the same daily climate step, for every station, before the HRUs use it (the recalculated value feeds the 30-day precipitation to PET ratio). Measured PET is therefore replaced by Hargreaves PET with a minimum of 0.01 mm, and `pet = 3` gives Hargreaves PET times `pet_co`. Channel and pond evaporation and aquifer revap use the same station value under every `pet` code.

With `pet = 3` the model also opens a file called `pet.cli` in the run folder and reads its first three lines. The file is not used after that.

Extra input columns

`event` is not an event switch. When it is not 0:

- each row of `plants.plt` is read with one more value at the end, the plant class (for example row crop, tree, grass). Calibration conditions can select plants by this class.
- each row of `soil_plant.ini` is read with one more name at the end, a constituent initial-condition record (see initial conditions).

Files written by the SWAT+ Editor do not have these columns, so `event` should be 0 for them.

Crack flow

With `crack = 1`, SWAT+ calculates a daily crack volume for each HRU from the soil's crack volume potential (`crk` in `soils.sol`) and the layer water contents. Surface runoff is reduced by the crack volume (in sub-daily runs, the runoff of each time step in turn until the volume is filled), and water entering the soil flows through the cracks before the layer-by-layer percolation.

Peak runoff rate

The peak runoff rate is used in the MUSLE sediment yield of each HRU.

With `sed_det = 0`, the peak rate comes from the NRCS method:

$$q_{peak} = \frac{\text{prf}}{6578.6} \cdot \frac{A\,Q_{surf}}{t_{conc}} \cdot \frac{1}{35.3}$$

where q_{peak} is in m³/s, A is the HRU area (ha), Q_{surf} is the surface runoff (mm), t_{conc} is the time of concentration (h) and `prf` is the peak rate factor in `parameters.bsn`. The factor 6578.6 converts ha · mm to mi² · in, and 35.3 converts ft³/s to m³/s. Wetlands always use this equation, whatever the value of `sed_det`.

With `sed_det = 1`, the half-hour rainfall intensity method is used:

$$\alpha_{tc} = 1 - \exp[2\,t_{conc}\,\ln(1-\alpha_{0.5})]\,, \qquad q_{peak} = \frac{\alpha_{tc}\,Q_{surf}\,A_{km}}{3.6\,t_{conc}}$$

where $\alpha_{0.5}$ is the day's fraction of rain falling in the maximum half hour (from the weather generator, scaled by `adj_pkrt` in `parameters.bsn`) and A_{km} is the HRU area in km².

Channel routing

`rte_cha` selects how water is routed through channels.

With `rte_cha = 0` (variable storage), the fraction of channel storage that leaves the reach in each routing step is

$$C_{sto} = \min\left(1,\ \text{scoef}\cdot\frac{2\,\Delta t}{2\,TT + \Delta t}\right)$$

where Δt is the routing step (h), TT is the travel time at the outflow of the previous step (h) and `scoef` is the channel storage coefficient in `parameters.bsn`. Daily runs use one routing step per day.

With `rte_cha = 1` (Muskingum), the outflow of a step is $O_2 = C_1 I_2 + C_2 I_1 + C_3 O_1$. The storage time constant is a weighted mean of the values at bankfull and at 0.1 bankfull depth, with the weights `msk_co1` and `msk_co2` (scaled to sum to 1), and `msk_x` is the weighting factor; all three are in `parameters.bsn`. The day is divided into enough substeps to keep the scheme stable.

USLE C factor

`c_fact` is replaced by 1 each time the C factor is calculated, so the model always uses the method based on residue and growing biomass cover (from APEX). The residue part of that method uses `rsd_cover` in `parameters.bsn`. The older method based on the plants' minimum C factor is not used in 62.0.1.

Carbon model

`carbon` selects the soil organic matter model:

- `0`: static carbon. Soil organic matter is kept in active and stable humus pools and in residue, with the original SWAT residue decomposition, mineralization and annual biological mixing.
- `2`: dynamic carbon model with CENTURY litter, microbial and humus pools. The model requires `carbon.bsn` and `carbon_lyr.bsn` and stops with an error if either is missing. Tillage, fertilizer and manure applications, grazing and the daily soil processes use the carbon pool versions of their routines, and the carbon outputs selected in `print.prt` become available (see carbon outputs).
- `1`: reserved for C-FARM, which is not implemented. No organic matter routine runs under this code: the soil humus pools are not initialised, and residue decomposition and N and P mineralization are skipped for the whole run.

The numbering changed in revision 62. The dynamic carbon model used to be selected with `carbon = 1`. A `codes.bsn` from an older dataset that still has `carbon = 1` now runs with no organic matter cycling, so it must be changed to `2` (dynamic model) or `0` (static model). See soil carbon for the model description.

Lapse rates

With `lapse = 1`, each HRU's daily precipitation and temperatures are adjusted for the difference between the HRU's elevation and the elevation of the gage or weather generator station that supplies the data:

$$P' = \max\left(0,\ P + \text{plaps}\cdot\frac{z_{HRU} - z_{gage}}{1000}\right), \qquad T' = T + \text{tlaps}\cdot\frac{z_{gage} - z_{HRU}}{1000}$$

with elevations in m, `plaps` in mm/km and `tlaps` in °C/km from parameters.bsn. The precipitation adjustment is a fixed depth added on every day, wet or dry, so with a positive `plaps` an HRU above its gage also receives rain on dry days. The temperature adjustment applies to the maximum, minimum and mean temperature. Reservoirs and channels use the station weather without adjustment.

Unit hydrograph

In sub-daily runs (more than one time step per day in time.sim) with the curve number method (`gampt = 0`), daily surface runoff from HRUs and routing units is spread over the time steps with a unit hydrograph. The base time and time to peak are

$$t_b = \min(48,\ 0.5 + 0.6\,t_{conc} + \text{adj_uhyd})\,, \qquad t_p = 0.375\,t_b$$

in hours, where `adj_uhyd` is in parameters.bsn. With `uhyd = 0` the hydrograph is triangular, rising to the peak at t_p and falling to zero at t_b. With `uhyd = 1` it follows the gamma function

$$q(t) = \left(\frac{t}{t_p}\right)^{\alpha} \exp\left[\alpha\left(1 - \frac{t}{t_p}\right)\right]$$

with α = `uhyd_alpha` from parameters.bsn. The ordinates are scaled to a total of 1 and cover at most two days. `uhyd` has no effect in daily runs or with Green-Ampt.

Runoff method

With `gampt = 0`, daily surface runoff is calculated with the curve number method. With `gampt = 1`, infiltration and runoff are calculated with the Green-Ampt method from the rainfall of each time step, which is intended for sub-daily runs. In sub-daily runs with Green-Ampt the runoff hydrograph of a routing unit is shifted in time without attenuation, and `uhyd` is not used. The shift is a fixed 2 hours for every routing unit; the computed time of concentration is not used for it. For urban HRUs, Green-Ampt also uses `urb_init_abst` in parameters.bsn.

Any `gampt` value above 0 also switches on the sub-daily erosion calculation from raindrop splash and overland flow, which uses `splash`, `rill`, `surq_exp` and `cov_mgt` in parameters.bsn.

In-stream nutrients

Channel water quality is always calculated with the QUAL2E routine. With `i_fpwet = 1`, simplified first-order transformations of organic N and sediment-bound P follow, using each channel's N and P transformation rates and the travel time, and their results replace the QUAL2E results for those constituents.

gwflow module

With `gw_flow = 1`, SWAT+ looks for chancell.gw in the run folder. If it is present and has data, the gridded groundwater module replaces the aquifer objects: the aquifer connection file is set to `null` and the groundwater cells connected to channels are added as objects. If chancell.gw is missing or empty, `gw_flow` is set back to 0 and the aquifer objects are used. See gwflow and object.cnt.

Tillage effect on decomposition

`idc_till` is used only with `carbon = 2`. It sets the tillage factor that multiplies the decomposition rates of the carbon pools in each soil layer:

idc_till	Tillage factor
1	DSSAT method: 1.6 in the tilled layers for till_eff_days days after tillage (from carbon.bsn), with a linear transition in a partly tilled layer; 1.0 otherwise.
2	Placeholder for the EPIC method. No tillage effect (factor 1.0).
3	Kemanian method: 1 plus the layer's tillage factor from tillage and biological mixing, calculated with the mixing curves in carbon.bsn.
4	Placeholder for the DNDC method. No tillage effect (factor 1.0).

Example

refdata/Ames_sub1/codes.bsn in the SWAT+ repository:

codes.bsn:

pet_file	wq_file	pet	event	crack	swift_out	sed_det	rte_cha	deg_cha
null	null	2	0	0	0	0	0	0

This file selects Hargreaves PET (pet = 2), the NRCS peak rate method, variable storage channel routing, curve number runoff, the dynamic carbon model (carbon = 2) and the Kemanian tillage method (idc_till = 3). cn = 2 and atmo_dep = 0 are in unused columns and have no effect. uhyd = 1 would select a gamma unit hydrograph if the run were sub-daily.

Related files

- file.cio: names codes.bsn on its basin line.
- parameters.bsn: coefficients used by the methods selected here.
- carbon.bsn and carbon_lyr.bsn: required when carbon = 2.
- hydrology.hyd: pet_co multiplies the PET of every method.
- Climate files: weather-sta.cli and pet.cli for station PET.
- gwflow: inputs of the gridded groundwater module.
- Soil carbon, nutrient cycling and channels: process descriptions.

Sources

- Reader: basin_read_cc.f90, called from proc_bsn.f90
- Type definition: type basin_control_codes in basin_module.f90
- File name slot: in_basin%codes_bas in input_file_module.f90, read in readcio_read.f90
- PET: et_pot.f90; station PET in climate_control.f90
- event: plant_parm_read.f90, soil_plant_init.f90
- Crack flow: sq_crackvol.f90, sq_crackflow.f90, swr_percmacro.f90, swr_percmain.f90
- SWIFT output: annual sums in command.f90, files written by swift_output.f90, called at the end of main.f90.in
- Peak rate and erosion: ero_pkq.f90, ero_cfactor.f90, ero_ovrsed.f90, surface.f90
- Channel routing: ch_rtmusk.f90, sd_hydsed_init.f90; in-stream nutrients in sd_channel_control3.f90
- Plant stress: pl_biomass_gro.f90
- Carbon model branches: hru_control.f90, soil_nutcarb_init.f90, carbon_bsn_read.f90, cbn_zhang2.f90
- Lapse rates: cli_lapse.f90, applied in hru_control.f90
- Unit hydrograph: unit_hyd.f90, unit_hyd_ru_hru.f90, hru_hyds.f90
- Runoff method: sq_volq.f90
- Tile drainage and water table: swr_percmain.f90
- gwflow activation: basin_read_objs.f90

Column	Fortran variable
pet_file	bsn_cc%petfile
wq_file	bsn_cc%wwqfile
pet	bsn_cc%pet

Column	Fortran variable
event	bsn_cc%nam1
crack	bsn_cc%crk
swift_out	bsn_cc%swift_out
sed_det	bsn_cc%sed_det
rte_cha	bsn_cc%rte
deg_cha	bsn_cc%deg
wq_cha	bsn_cc%wq
nostress	bsn_cc%nostress
cn	bsn_cc%cn
c_fact	bsn_cc%cfac
carbon	bsn_cc%cswat
lapse	bsn_cc%lapse
uhyd	bsn_cc%uhyd
sed_cha	bsn_cc%sed_ch
tiledrain	bsn_cc%tdrn
wtable	bsn_cc%wtdn
soil_p	bsn_cc%sol_p_model
gampt	bsn_cc%gampt
atmo_dep	bsn_cc%atmo
stor_max	bsn_cc%smax
i_fpwet	bsn_cc%qual2e
gw_flow	bsn_cc%gwflow
idc_till	bsn_cc%idc_till

Purpose

`parameters.bsn` contains the coefficients that apply to the whole watershed: initial soil water, surface runoff lag, peak rate factors, nutrient and pesticide transport coefficients, denitrification, channel routing coefficients, lapse rates, erosion coefficients for sub-daily runs and the starting CO2 concentration. It is named in the second position of the `basin` line of `file.cio` (default `parameters.bsn`). Several of the parameters only matter for the methods selected in `codes.bsn`.

If the file named in `file.cio` does not exist, every parameter keeps its built-in value, after the zero-value replacements described in Zero and missing values.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Line 3 contains 44 values in the order below: 42 real numbers followed by two integers. The reader matches values by position only, so all 44 must be present and in this order. The table gives the column headers the SWAT+ Editor writes; several no longer describe what the model does with the value.

#	Column	Units	Built-in value	Typical range	Description
1	`lai_noevap`	-	3.0	-	Not used. Read and can be changed by calibration, but no routine uses it.
2	`sw_init`	fraction	0.0	0 to 1	Initial soil water as a fraction of field capacity, the same for all soils. 0 means it is calculated from average annual precipitation. See Initial soil water.
3	`surq_lag`	-	4.0	1 to 24	Surface runoff lag coefficient. See Surface runoff lag.
4	`adj_pkrt`	-	1.0	0.5 to 2	Multiplier on the monthly mean fraction of daily rain that falls in the maximum half hour, calculated from the weather generator statistics (limited to 0.1 to 0.95). The daily fraction drawn from it is used by the half-hour intensity peak rate method (`sed_det = 1`), the rainfall erosivity of the MUSLE, urban washoff and the generation of sub-daily rainfall.
5	`adj_pkrt_sed`	-	484.0	-	Peak rate factor of the NRCS peak runoff equation, used for HRU erosion when `sed_det = 0` in `codes.bsn` and for wetlands. See Peak rate factor.
6	`lin_sed`	-	0.0	0 and up	Pesticide groundwater mixing factor. See Pesticides in aquifers.
7	`exp_sed`	-	0.0	0 or 1	Temperature-dependent pesticide decay: `1` on, any other value off. See Pesticide decay and temperature.
8	`orgn_min`	1/day	0.003	0.0003 to 0.003	Rate factor for mineralization of the active humus N and P pools (static carbon model, `carbon = 0`).
9	`n_uptake`	-	20.0	0 to 100	Nitrogen uptake distribution parameter. See Uptake distribution.
10	`p_uptake`	-	20.0	0 to 100	Phosphorus uptake distribution parameter.
11	`n_perc`	-	0.10	0 to 1	Nitrate percolation coefficient: ratio of the nitrate concentration in surface runoff to the concentration in water leaving the top layer.
12	`p_perc`	m^3/Mg	10.0	10 to 17.5	Phosphorus percolation coefficient, used in the leaching of soluble P from each layer.
13	`p_soil`	m^3/Mg	175.0	100 to 200	Phosphorus soil partitioning coefficient: ratio of soluble P in the top 10 mm of soil to soluble P in surface runoff.

#	Column	Units	Built-in value	Typical range	Description
14	p_avail	fraction	0.40	0.01 to 0.7	Phosphorus availability index: fraction of fertilizer P that is in solution after the fast reaction period. Sets the balance between the labile and active mineral P pools in the original P model (soil_p = 0).
15	rsd_decomp	fraction/day	0.05	0.02 to 0.1	Residue decomposition coefficient: daily fraction of residue that decomposes under ideal conditions. Used for every plant whose plnt_decomp in plants.plt is 0.
16	pest_perc	-	0.5	0 to 1	Pesticide percolation coefficient: ratio of the pesticide concentration in surface runoff and lateral flow to the concentration in percolate.
17	msk_co1	-	0.75	0 to 10	Weight of the storage time constant at bankfull depth in the Muskingum storage constant.
18	msk_co2	-	0.25	0 to 10	Weight of the storage time constant at 0.1 bankfull depth in the Muskingum storage constant.
19	msk_x	-	0.20	0 to 0.3	Muskingum weighting factor for inflow and outflow in channel storage.
20	nperco_lchtile	-	0.5	0 to 1	Ratio of the nitrate concentration in tile flow to the concentration in the mobile water of the tile layer.
21	evap_adj	-	0.60	0.5 to 1	Channel evaporation adjustment factor: multiplier on station PET for evaporation from the channel water surface.
22	scoef	-	1.0	0 to 1	Channel storage coefficient of the variable storage routing method (rte_cha = 0 in codes.bsn).
23	denit_exp	-	1.40	0 to 3	Denitrification exponential rate coefficient. See Denitrification.
24	denit_frac	-	1.30	see text	Denitrification threshold on the soil water factor. See Denitrification.
25	man_bact	fraction	0.15	-	Not used.
26	adj_uhyd	h	0.0	0 and up	Added to the base time of the sub-daily unit hydrograph. See unit hydrograph.
27	cn_froz	1/mm	0.000862	-	Frozen soil coefficient of the curve number retention parameter. See Frozen soil.

#	Column	Units	Built-in value	Typical range	Description
28	dorm_hr	h	-1.0	0 to 24	Day length threshold for dormancy. 0 or less means it is calculated from latitude. See Dormancy threshold.
29	plaps	mm/km	0.0	-	Precipitation lapse rate. Positive values increase precipitation with elevation. Used when lapse = 1 in codes.bsn.
30	tlaps	°C/km	6.5	-	Temperature lapse rate. Positive values decrease temperature with elevation. Used when lapse = 1 in codes.bsn.
31	n_fix_max	kg N/ha/day	20.0	1 to 20	Maximum daily nitrogen fixation of a plant.
32	rsd_decay	fraction/day	0.01	0 to 0.05	Minimum daily residue decay fraction. The daily decay rate of residue is never below this value.
33	rsd_cover	ha/t	0.75	-	Residue cover factor in the USLE C factor. See USLE C factor.
34	urb_init_abst	mm	1.0	0 to 10	Maximum initial abstraction on impervious urban areas. Used only with Green-Ampt (gampt = 1).
35	petco_pmpt	%	100.0	-	Not used. The PET of every method is adjusted with pet_co in hydrology.hyd instead.
36	uhyd_alpha	-	1.0	0.5 to 10	Shape coefficient of the gamma unit hydrograph (uhyd = 1 in codes.bsn).
37	splash	-	0.0	0.9 to 3.1	Splash erosion coefficient. See Sub-daily erosion.
38	rill	-	0.0	0.5 to 2	Rill erosion coefficient, a multiplier on the USLE K factor of the top layer.
39	surq_exp	-	0.0	1 to 3	Exponent on bed shear stress in overland flow erosion.
40	cov_mgt	-	0.0	0.001 to 0.45	Scaling factor on the cover and management factor in overland flow erosion.
41	cha_d50	mm	0.0	-	Not used. The median particle size of each channel is read from the channel input.
42	co2	ppm	400.0	-	Atmospheric CO2 concentration, used for every year when there is no co2_yr.dat. See CO2.
43	day_lag_max	days	0	-	Not used as read. The model always sets it to 2.
44	igen	-	5	0 and up	Random number seed code. See Random numbers.

Values are read in one free-format read. A line with fewer values continues onto the next line, and the parameters that are not reached keep their built-in values. A value that cannot be read ends the read without a message.

Zero and missing values

After reading, SWAT+ replaces values that are zero or negative (strictly, below 10^{-6}) for some parameters. For most of them the replacement is the built-in value, so leaving a column at 0 has the same effect as entering the built-in value. These are: `lai_noevap`, `surq_lag`, `adj_pkrt`, `n_uptake`, `p_uptake`, `n_perc`, `p_perc`, `p_soil`, `p_avail`, `rsd_decomp`, `pest_perc`, `msk_co1`, `msk_co2`, `msk_x`, `nperco_lchtile`, `evap_adj`, `denit_exp`, `denit_frac`, `man_bact`, `cn_froz`, `n_fix_max`, `rsd_decay`, `rsd_cover` and `uhyd_alpha`.

The exceptions are:

Column	Value read	Value used
`adj_pkrt_sed`	0 or less	1.0 (the built-in value is 484.0)
`orgn_min`	0 or less	0.0003 (the built-in value is 0.003)
`urb_init_abst`	0 or less	0.0 (the built-in value is 1.0)
`co2`	below 100	400.0
`day_lag_max`	any	2

`sw_init`, `adj_uhyd`, `dorm_hr`, `splash`, `rill`, `surq_exp`, `cov_mgt` and `cha_d50` stay at 0 when 0 is read, and 0 has a meaning for some of them (see the sections below). `lin_sed`, `exp_sed`, `scoef`, `plaps`, `tlaps` and `igen` are used as read.

Initial soil water

All soils start at the same fraction of field capacity. If `sw_init` is 0 or less, the fraction is calculated from the average annual precipitation P (mm) of the HRU's weather generator station:

$$f_{fc} = \frac{P}{P + \exp(9.043 - 0.002135\,P)}$$

A warm-up period of at least one year (`nyskip` in `print.prt`) makes results largely independent of `sw_init`.

Surface runoff lag

In HRUs with a long time of concentration, only part of the surface runoff reaches the channel on the day it is generated. The fraction released each day is

$$f_{rel} = 1 - \exp\left(-\frac{\text{surq_lag}}{t_{conc}}\right)$$

with t_{conc} in hours. In sub-daily runs t_{conc} is expressed in time steps. The rest is kept in surface storage and added to the next day's runoff. Smaller values of `surq_lag` keep more water in storage and smooth the hydrograph.

Peak rate factor

The column labelled `adj_pkrt_sed` by the SWAT+ Editor is the peak rate factor of the NRCS peak runoff equation (see peak runoff rate). The built-in value, 484, is the standard factor for the NRCS dimensionless unit hydrograph. The Editor writes 1.0 in this column, and a value of 0 is also replaced by 1.0. A factor of 1 gives peak rates 484 times lower than the standard factor, which lowers MUSLE sediment yields from HRUs when `sed_det = 0`. The value only affects the peak rate used for erosion and in wetlands; it does not change runoff volumes.

Pesticides in aquifers

The columns labelled `lin_sed` and `exp_sed` by the Editor are pesticide parameters in 62.0.1. In each aquifer, the pesticide concentration in return flow and seepage is the pesticide mass divided by a mixing volume:

$$C = \frac{M_{pest}}{f_{mix}\, S_{aqu} + Q_{rtn} + Q_{revap} + Q_{seep}}$$

where f_{mix} is `lin_sed`, S_{aqu} is the aquifer storage and the Q terms are the day's return flow, revap and seepage (mm). With `lin_sed = 0` the concentration is based on the day's outflows only. Larger values dilute the pesticide in more of the stored water and slow its release.

Pesticide decay and temperature

With `exp_sed = 1` (the value 1.0; other values such as 2 do not count), pesticide decay in soil, on foliage, in aquifers, in channel water and in channel bed sediment uses half-lives adjusted for temperature with a Q10 model:

$$t_{1/2}(T) = t_{1/2,25}\, Q_{10}^{(25-T)/10}, \qquad Q_{10} = 2.58$$

where $t_{1/2,25}$ is the half-life from the pesticide database, taken as the value at 25 °C, and the daily fraction remaining is $\exp(-0.693/t_{1/2}(T))$. Soil decay uses the layer temperature, foliar decay the top soil layer temperature, and channel decay the water temperature. In aquifers the temperature is fixed at 0 °C, which makes aquifer half-lives about 10.7 times longer than the database value. Any other value of `exp_sed` uses the database half-lives without a temperature adjustment.

The Ames example file (see Example) has `exp_sed = 1.0`, so temperature-dependent decay is on in that dataset.

Uptake distribution

The maximum nitrogen uptake from the soil above depth z is

$$U(z) = U_{pot}\, \frac{1 - \exp(-\beta_n\, z / z_{root})}{1 - \exp(-\beta_n)}$$

where β_n is `n_uptake` and z_{root} is the root depth. Larger values concentrate uptake near the surface. Because the top 10 mm interacts with surface runoff, `n_uptake` affects the nitrate available for runoff; lower layers can make up a shortfall in the upper layers, so plant N stress changes little with this parameter. `p_uptake` works the same way for phosphorus.

Denitrification

Denitrification in a soil layer is

$$N_{denit} = NO_3 \left[1 - \exp\left(-\text{denit_exp}\ \gamma_{tmp}\ \frac{OC}{100}\right)\right]$$

where γ_{tmp} is the temperature factor and OC the organic carbon content (%). It only occurs on days when the soil water factor γ_{sw} is at or above `denit_frac`.

With the static carbon model (`carbon = 0`), $\gamma_{sw} = 0.1 + 0.9\sqrt{SW/FC}$, with no upper limit. With the default `denit_frac` of 1.3, denitrification needs $SW/FC \geq 1.78$, that is, a layer well above field capacity.

With the dynamic carbon model (`carbon = 2`), γ_{sw} comes from the water factor selected in `carbon.bsn` and is never above 1. Any `denit_frac` above 1, including the default 1.3, then switches denitrification off for the whole run. Datasets using `carbon = 2` need a `denit_frac` of 1 or less to simulate denitrification.

Frozen soil

When the second soil layer is at or below 0 °C, the curve number retention parameter S is reduced:

$$S_{frz} = S_{max}\,[1 - \exp(-\text{cn_froz} \cdot S)]$$

which increases runoff from frozen soil. The same adjustment is applied with Green-Ampt.

Dormancy threshold

A plant goes dormant when the day length falls below the station's minimum day length plus a threshold t_{dorm} (hours). If `dorm_hr` is above 0, t_{dorm} = `dorm_hr` everywhere. Otherwise it depends on the latitude ϕ of the weather generator station: 1 h above 40°, $(|\phi| - 20)/20$ h between 20° and 40°, and -1 h below 20°.

USLE C factor

The C factor is the product of a residue factor and a biomass factor:

$$C = \exp\left(-\text{rsd_cover} \cdot \frac{R+1}{1000}\right)\,[1 - f_{cov}\,\exp(-0.328\,h_c)]$$

where R is the residue on the surface (kg/ha), h_c is the canopy height (m), and f_{cov} is the ground cover fraction of the above-ground biomass. This is the only C factor method in 62.0.1 (see USLE C factor).

Sub-daily erosion

With Green-Ampt (`gampt` above 0 in `codes.bsn`), sediment from each time step is the sum of splash and overland flow erosion. Splash erosion decreases with runoff depth h (mm) as $\exp(-\text{splash} \cdot h/1000)$ and depends on rainfall kinetic energy and a soil texture erodibility. Overland flow erosion is

$$E_{ov} = 11.02\ \text{rill}\ K\ \text{cov_mgt}\ C\ \tau^{\,\text{surq_exp}}$$

in kg/h/m², where K is the USLE K factor of the top layer, C is a cover factor from biomass and residue, and τ is the bed shear stress (N/m²). With the built-in values of 0 for `rill` or `cov_mgt`, overland flow erosion is zero.

CO2

Without a `co2_yr.dat` file in the run folder, every year uses `co2`. With the file, the yearly values come from it (see co2_yr.dat). The radiation use efficiency of plants is adjusted for CO2 only when `co2` in `parameters.bsn` is above 350 ppm, even when the yearly values come from `co2_yr.dat`. Penman-Monteith PET uses the yearly values in all cases.

Random numbers

With `igen = 0`, the weather generator and other random processes start from fixed seeds, so every run gives the same random sequence. With any other value, each seed is advanced by a number of steps that depends on `igen`, so different `igen` values give different sequences. The same `igen` always gives the same sequence, so a new value is needed for each different realisation.

Calibration

These parameters can be changed during calibration through calibration.cal with object type bsn, using their model names: surlag (surq_lag), adj_pkr (adj_pkrt), prf (adj_pkrt_sed), evlai (lai_noevap), ffcb (sw_init), cmn (orgn_min), nperco (n_perc), pperco (p_perc), phoskd (p_soil), psp (p_avail), rsdco (rsd_decomp), percop (pest_perc), msk_co1, msk_co2, msk_x, nperco_lchtile, evrch (evap_adj), cdn (denit_exp), sdnco (denit_frac), tb_adj (adj_uhyd), n_updis (n_uptake), rsd_covco (rsd_cover), dorm_hr, plaps and tlaps. See parameter change files.

The changes in calibration.cal are applied once, after the HRUs, channels and weather stations have been set up. Several of these parameters are only used during that setup, so changing them in calibration.cal has no effect on the run:

- cmn, nperco, pperco, phoskd, psp and nperco_lchtile have already been copied to each HRU. Use the HRU-level names cmn_h, nperco_h, pperco_h, phoskd_h, psp_h and nperco_lchtile_h instead.
- ffcb: the initial soil water has already been set.
- adj_pkr and dorm_hr: the half-hour rainfall fractions and the dormancy thresholds of the weather stations have already been calculated.
- msk_co1, msk_co2 and msk_x: the Muskingum coefficients of the channels have already been calculated.
- rsdco: it has already been copied to the plants with plnt_decomp = 0.

surlag is recalculated into the runoff release fraction when it is changed, and the others are used directly during the simulation, so their changes take effect.

Example

refdata/Ames_sub1/parameters.bsn in the SWAT+ repository:

```
parameters.bsn: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
  lai_noevap       sw_init      surq_lag      adj_pkrt  adj_pkrt_sed       lin_sed       exp_sed      orgn
     3.00000       0.50000       4.00000       1.00000       1.00000       0.00000       1.00000       0.0
```

How the model reads some of these values:

- sw_init = 0.5: soils start at half of field capacity.
- adj_pkrt_sed = 1.0: the NRCS peak rate factor is 1. The standard factor is 484.
- exp_sed = 1.0: temperature-dependent pesticide decay is on.
- orgn_min = 0.0: replaced by 0.0003.
- cn_froz = 0.0: replaced by 0.000862.
- dorm_hr = 0.0: the dormancy threshold is calculated from latitude.
- plaps = 0.0 and tlaps = 0.0: no lapse rate adjustment, even if lapse = 1 in codes.bsn.
- denit_frac = 1.3: this dataset uses carbon = 2 in its codes.bsn, so no denitrification occurs (see Denitrification).
- petco_pmpt, cha_d50 and man_bact have no effect; day_lag_max is set to 2.
- igen = 0: fixed random seeds.

Related files

- file.cio: names parameters.bsn on its basin line.
- codes.bsn: method codes that decide which of these parameters are used.
- hydrology.hyd: HRU-level hydrology parameters, including pet_co.
- plants.plt: plnt_decomp overrides rsd_decomp for a plant.
- Parameter change files: calibration names for these parameters.
- Nutrient cycling, pesticides, erosion and sediment, channels and hydrology: process descriptions.

Sources

- Reader: basin_read_prm.f90, followed by the zero-value replacements in basin_prm_default.f90; both called from proc_bsn.f90

- Type definition: `type basin_parms` in basin_module.f90
- File name slot: `in_basin%parms_bas` in input_file_module.f90
- Copy to HRUs: hrudb_init.f90
- Hydrology: hydro_init.f90, time_conc_init.f90, sq_dailycn.f90, sq_greenampt.f90, unit_hyd.f90, cli_lapse.f90
- Weather generator: cli_initwgn.f90, gcycl.f90, co2_read.f90
- Peak rate and erosion: ero_pkq.f90, wetland_control.f90, ero_cfactor.f90, ero_ovrsed.f90
- Pesticides: aqu_1d_control.f90, pest_decay.f90, pest_tempdecay_module.f90, ch_rtpest.f90, pest_lch.f90
- Nutrients: nut_nminrl.f90, cbn_zhang2.f90, nut_nlch.f90, nut_solp.f90, nut_pminrl.f90, rsd_decomp.f90, cbn_surfrsd_decomp.f90, plantparm_init.f90
- Plants: pl_nup.f90, pl_pup.f90, pl_nfix.f90, pl_biomass_gro.f90
- Channels: sd_hydsed_init.f90, ch_rtmusk.f90, ch_rthr.f90, ch_rtday.f90
- Calibration: cal_parm_select.f90

Column	Fortran variable
lai_noevap	bsn_prm%evlai
sw_init	bsn_prm%ffcb
surq_lag	bsn_prm%surlag
adj_pkrt	bsn_prm%adj_pkr
adj_pkrt_sed	bsn_prm%prf
lin_sed	bsn_prm%pestgwfact
exp_sed	bsn_prm%temp_decay
orgn_min	bsn_prm%cmn
n_uptake	bsn_prm%n_updis
p_uptake	bsn_prm%p_updis
n_perc	bsn_prm%nperco
p_perc	bsn_prm%pperco
p_soil	bsn_prm%phoskd
p_avail	bsn_prm%psp
rsd_decomp	bsn_prm%rsdco
pest_perc	bsn_prm%percop
msk_co1, msk_co2, msk_x	bsn_prm%msk_co1, %msk_co2, %msk_x
nperco_lchtile	bsn_prm%nperco_lchtile
evap_adj	bsn_prm%evrch
scoef	bsn_prm%scoef
denit_exp	bsn_prm%cdn
denit_frac	bsn_prm%sdnco
man_bact	bsn_prm%bact_swf
adj_uhyd	bsn_prm%tb_adj
cn_froz	bsn_prm%cn_froz
dorm_hr	bsn_prm%dorm_hr
plaps, tlaps	bsn_prm%plaps, %tlaps
n_fix_max	bsn_prm%nfixmx
rsd_decay	bsn_prm%decr_min
rsd_cover	bsn_prm%rsd_covco
urb_init_abst	bsn_prm%urb_init_abst
petco_pmpt	bsn_prm%petco_pmpt
uhyd_alpha	bsn_prm%uhalpha
splash	bsn_prm%eros_spl
rill	bsn_prm%rill_mult
surq_exp	bsn_prm%eros_expo
cov_mgt	bsn_prm%c_factor
cha_d50	bsn_prm%ch_d50
co2	bsn_prm%co2
day_lag_max	bsn_prm%day_lag_mx
igen	bsn_prm%igen

Purpose

carbon.bsn contains the basin-wide parameters of the dynamic soil carbon model (carbon = 2 in codes.bsn): how soil organic carbon is split into pools at the start of the run, losses of soluble carbon in water, the split of organic N and P in fertilizer and manure, the duration and decay of tillage effects, the curves for tillage and biological mixing, the temperature and water factors on decomposition, surface residue photodegradation, and the C:N and C:P controls on residue decomposition.

It works together with carbon_lyr.bsn, which contains the decomposition rates and CO2 fractions for the surface layer and for the deeper layers. In revision 62 these two files replaced carb_coefs.cbn (see carb_coefs.cbn).

Activation and file names

The file is read only when carbon = 2 in codes.bsn. With any other carbon code it is ignored and need not exist.

With carbon = 2 the file is required. SWAT+ writes an error to the screen and to diagnostics.out and stops if:

- the file does not exist or cannot be opened;
- its data line is missing or one of the 28 values cannot be read;
- the companion file carbon_lyr.bsn is missing or has too few valid rows (see carbon_lyr.bsn).

There are no built-in values to fall back on.

The file is named in the third position of the basin line of file.cio, after codes.bsn and parameters.bsn. The default name is carbon.bsn. The name of the layer file is built from it by inserting _lyr before the extension: carbon.bsn gives carbon_lyr.bsn, mybasin.bsn gives mybasin_lyr.bsn, and a name without an extension gets _lyr.bsn added. The layer file is not listed in file.cio.

The basin line of file.cio must have three names. The line is read as three names in one free-format read, so a basin line from an older dataset with only two names takes the first word of the next line (normally climate) as the carbon file name, and the rest of file.cio is then read one line out of step. Datasets that do not use the carbon model can put null in the third position.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped and may be blank. Line 3 contains the 28 values in the order below. The reader matches values by position only. Columns 7, 11, 12 and 28 are integers and must be written without a decimal point, for example 2; a value such as 2.0 in these columns stops the run with a read error.

#	Column	Units	Example value	Description
1	init_seq	fraction	0.95	Fraction of each layer's soil organic carbon put into the humus and microbial pools at the start. The rest, 1− init_seq, goes to metabolic (85%) and structural (15%) litter.
2	init_microb	fraction	0.02	Fraction of the init_seq share that goes to the microbial biomass pool.
3	init_slow	fraction	0.44	Fraction of the init_seq share that goes to the slow humus pool (when mathers_method is 0).
4	init_passive	fraction	0.54	Fraction of the init_seq share that goes to the passive humus pool.
5	koc_c	-	1000	Partition coefficient of microbial carbon between soil and water, controlling its loss in runoff, lateral flow and percolation from the top layer. Typical range 500 to 1500.

#	Column	Units	Example value	Description
6	`solc_ratio`	-	0.5	Ratio of the soluble carbon concentration in surface runoff and lateral flow to the concentration in percolate. Typical range 0.1 to 1.
7	`till_eff_days`	days (integer)	100	Number of days a tillage operation affects decomposition.
8	`manure_c_frac`	fraction	0.5	Split of the organic N and P in applied fertilizer and manure between fresh residue and the slow humus pool. See Organic fertilizer and manure.
9	`bio_consol`	fraction/day	0.15	Daily recovery rate of biological mixing after tillage.
10	`till_consol`	fraction/day	0.10	Daily decline of the tillage factor as the soil settles.
11	`tmpf_eqn`	code (integer)	2	Temperature factor equation: `1`, `2` or `3`. See Temperature factor.
12	`watf_eqn`	code (integer)	1	Water factor equation: `1` or `2`. See Water factor.
13	`t_cbn_min`	°C	-0.5	Minimum temperature of temperature equation 2.
14	`t_cbn_opt`	°C	30.0	Optimum temperature of temperature equation 2.
15	`t_cbn_max`	°C	50.0	Maximum temperature of temperature equation 2.
16	`bmix_a`	-	3.0	Intercept a of the texture curve for biological mixing.
17	`bmix_b`	-	5.0	Coefficient b of the texture curve for biological mixing.
18	`bmix_c`	-	-5.5	Exponent coefficient c of the texture curve for biological mixing.
19	`tillmix_a`	-	3.0	Intercept a of the texture curve for tillage.
20	`tillmix_b`	-	15.0	Coefficient b of the texture curve for tillage.
21	`tillmix_c`	-	-3.5	Exponent coefficient c of the texture curve for tillage.
22	`sfc_rsd_photodeg`	fraction/day	0.001	Fraction of surface residue lost each day to photodegradation. The lost carbon is counted as a residue emission.
23	`n_act_frac`	fraction	0.02	Fraction of organic N in the active humus pool. Not used with `carbon = 2`. See Notes on individual columns.
24	`cnr_cap`	-	500	Upper limit on the residue C:N ratio in the C:N decomposition factor.
25	`cnr_ref`	-	25	Residue C:N ratio at which the C:N decomposition factor equals 1.
26	`cpr_cap`	-	5000	Upper limit on the residue C:P ratio in the C:P decomposition factor.
27	`cpr_ref`	-	200	Residue C:P ratio at which the C:P decomposition factor equals 1.
28	`mathers_method`	code (integer)	1	Slow humus initialisation: `1` Mathers method, any other value the `init_slow` fraction. See Initial carbon pools.

The example values are those of `refdata/Ames_sub1/carbon.bsn` (see Example). Since the file is required, there are no defaults in the usual sense.

Initial carbon pools

At the start of the run, the organic carbon C of each soil layer (from the organic carbon content in `soils.sol`) is divided into pools:

Pool	Carbon	C:N	C:P
passive humus	$C \cdot \text{init_seq} \cdot \text{init_passive}$	10	80
slow humus	$C \cdot \text{init_seq} \cdot \text{init_slow}$	10	80
microbial biomass	$C \cdot \text{init_seq} \cdot \text{init_microb}$	8	80
metabolic litter	$C \cdot (1 - \text{init_seq}) \cdot 0.85$	10	100
structural litter	$C \cdot (1 - \text{init_seq}) \cdot 0.15$	150	1500

The reader does not check the fractions. `init_microb`, `init_slow` and `init_passive` should add up to 1, or the pools will not add up to the soil's organic carbon.

With `mathers_method = 1`, the slow humus pool is set from the clay and silt content instead of `init_slow`, following "Updating carbon pool initialization with DSSAT-CENTURY". In 62.0.1 the test and equation are written for clay plus silt as a fraction, while the soil values are in percent. As a result, for every layer with more than 0.35% clay plus silt the slow humus share is

$$f_{slow} = 0.0041 + 0.0053\,(\text{clay} + \text{silt})$$

with clay and silt in percent (0.32 for a layer with 60% clay plus silt), and the slow humus carbon is $C \cdot \text{init_seq} \cdot f_{slow}$. The passive and microbial pools keep their `init_passive` and `init_microb` shares, so with the Mathers method the pools generally do not add up to the soil's organic carbon.

Soluble carbon losses

Each day, microbial carbon in the top layer is lost with the water leaving it:

$$C_{loss} = C_{mic}\left[1 - \exp\left(-\frac{Q_{surf} + Q_{perc} + Q_{lat}}{UL + 0.0001\,\text{koc_c} \cdot C_{tot}}\right)\right]$$

where UL is the water content of the layer at saturation (mm) and C_{tot} is the layer's total organic carbon. The loss is shared between percolation and surface runoff plus lateral flow, with the concentration in surface runoff and lateral flow equal to `solc_ratio` times the concentration in percolate. Microbial carbon attached to eroded sediment is lost in proportion to the sediment yield.

Organic fertilizer and manure

When fertilizer with organic N or P, or manure, is applied, the organic matter goes to the slow humus pool and to metabolic and structural litter. In the code, the slow humus pool receives the fraction 1− `manure_c_frac` of the organic matter in each layer, and the metabolic and structural litter together also receive 1− `manure_c_frac`, split by a metabolic fraction from the C:N ratio of the material. The code comments say litter should receive the fraction `manure_c_frac`. With the value 0.5 used in the reference datasets the two readings give the same result; with other values the amounts added differ from the comments.

Tillage and biological mixing

The tillage factor of a soil layer is calculated from a texture curve and the cumulative soil disturbance x of tillage operations:

$$Z = a + b\,\exp\left(c\,\frac{\text{clay}}{100}\right), \qquad F = Z\,\frac{x}{x + \exp(1 - 2x)}$$

with clay in percent. Tillage operations use `tillmix_a`, `tillmix_b` and `tillmix_c`; biological mixing uses `bmix_a`, `bmix_b` and `bmix_c`. The layer's tillage factor is the sum of the two parts. The original equations were $Z = 3 + 12\,\exp(-5.5\,\text{clay}/100)$ for tillage and $Z = 3 + 5\,\exp(-5.5\,\text{clay}/100)$ for biological mixing.

After tillage, the tillage part declines each day by the fraction `till_consol` (scaled down by SW/FC when the layer is drier than field capacity), and biological mixing efficiency recovers each day by `bio_consol` times its initial value until it is back to that value. `till_eff_days` days after a tillage operation the tillage effect ends and the tillage part is reset to 0.

How the tillage factor changes decomposition depends on `idc_till` in `codes.bsn`. With `idc_till = 3` the decomposition rates are multiplied by $1 + F$. With `idc_till = 1` they are multiplied by 1.6 in tilled layers for `till_eff_days` days.

Temperature factor

`tmpf_eqn` selects the temperature factor γ_{tmp} on decomposition, using the layer temperature T (°C). Decomposition only occurs when $T > 0$.

`tmpf_eqn`	Equation
1	$\gamma_{tmp} = T/[T + \exp(5.058 - 0.2504\,T)]$ for $T \le 35$; $1 - 0.04\,(T - 35)$ for $35 < T \le 60$; 0 above 60.
2	$\gamma_{tmp} = \dfrac{(T - T_n)^q\,(T_x - T)}{(T_o - T_n)^q\,(T_x - T_o)}$ with $q = \dfrac{T_n - T_o}{T_o - T_x}$, and 0 outside $T_n < T < T_x$. T_n, T_o and T_x are `t_cbn_min`, `t_cbn_opt` and `t_cbn_max`.
3	$\gamma_{tmp} = 0.9\,T/[T + \exp(9.93 - 0.312\,T)] + 0.1$

Biological mixing is also reduced with temperature: by equation 3 when `tmpf_eqn = 3`, and by equation 2 otherwise. So `t_cbn_min`, `t_cbn_opt` and `t_cbn_max` affect biological mixing even when `tmpf_eqn = 1`.

Water factor

`watf_eqn` selects the water factor γ_{sw}:

- `1`: $\gamma_{sw} = 0.1 + 0.9\sqrt{SW/FC}$, limited to 0.05 to 1, where SW and FC are the layer's water content and field capacity above wilting point.
- `2`: the product of a water term, rising from 0 at zero water content to 0.4 at wilting point and 1 at field capacity, and an aeration term that falls as the air-filled porosity drops below 0.1. Both terms are at most 1.

Decomposition rates are multiplied by the combined factor $\min\left(15,\ 0.9\,\sqrt{\gamma_{tmp}\,\gamma_{sw}}\,\gamma_{ox}\,f_{till}\right)$, where γ_{ox} is an oxygen factor that decreases with depth and f_{till} is the tillage factor.

With either equation γ_{sw} never exceeds 1. Denitrification in the dynamic carbon model occurs only when γ_{sw} reaches `denit_frac` in `parameters.bsn`, so a `denit_frac` above 1 (the default is 1.3) switches denitrification off.

Residue decomposition

The daily decay rate of surface residue for each plant is

$$k = \max\left(\text{rsd_decay},\ k_{pl}\ \min(f_{CN}, f_{CP}, 1)\ \sqrt{\gamma_{tmp}\,\gamma_{sw}}\right), \qquad f_{CN} = \exp\left(-0.693\,\frac{CN - \text{cnr_ref}}{\text{cnr_ref}}\right)$$

limited to 1, where k_{pl} is the plant's residue decomposition coefficient (`plnt_decomp` in `plants.plt`, or `rsd_decomp` in `parameters.bsn`), CN is the residue C:N ratio limited to `cnr_cap`, and f_{CP} is the same expression with the C:P ratio, `cpr_ref` and `cpr_cap`. A residue C:N ratio equal to `cnr_ref` gives $f_{CN} = 1$; each further `cnr_ref` halves it. Before decomposition, the fraction `sfc_rsd_photodeg` of the surface residue is removed by photodegradation.

Notes on individual columns

- `n_act_frac` is used only by the static carbon routines (`carbon = 0`), which do not read `carbon.bsn`. With `carbon = 2` the value read has no effect, and with `carbon = 0` the model uses 0.02.
- `till_eff_days` is the only tillage duration in the model. Under `idc_till = 3` it also limits how long the tillage part of the tillage factor lasts.
- `tmpf_eqn` and `watf_eqn` values other than those listed leave the factor at 0, which stops decomposition of the soil carbon pools and denitrification in every layer.

Calibration

These parameters can be changed during calibration through calibration.cal with object type `bsn`: `init_seq`, `init_microb`, `init_slow`, `init_passive`, `koc_c`, `solc_ratio`, `manure_c_frac`, `bio_consol`, `till_consol`, `sfc_rsd_photodeg`, `n_act_frac`, `cnr_cap`, `cnr_ref`, `cpr_cap`, `cpr_ref`, `t_cbn_min`, `t_cbn_opt`, `t_cbn_max`, `bmix_a`, `bmix_b`, `bmix_c`, `tillmix_a`, `tillmix_b` and `tillmix_c`. `till_eff_days`, `tmpf_eqn`, `watf_eqn` and `mathers_method` cannot be calibrated. A change to `n_act_frac` has no effect with `carbon = 2`. See parameter change files.

Example

`refdata/Ames_sub1/carbon.bsn` in the SWAT+ repository:

```
carbon.bsn: derived from carb_coefs.cbn (basin carbon scalars)
          init_seq       init_microb         init_slow      init_passive             koc_c        solc_rat
              0.95              0.02              0.44              0.54            1000.0               0
```

95% of soil organic carbon starts in the humus and microbial pools. Because `mathers_method = 1`, the slow humus pool is set from clay and silt, and `init_slow` is not used. Temperature equation 2 runs from -0.5 °C through an optimum of 30 °C to 50 °C, and the water factor is equation 1. Tillage affects decomposition for 100 days.

Related files

- carbon_lyr.bsn: decomposition rates and CO2 fractions for the surface and deeper layers.
- codes.bsn: `carbon = 2` turns the carbon model on; `idc_till` selects the tillage method.
- file.cio: names `carbon.bsn` on its `basin` line.
- parameters.bsn: `rsd_decomp`, `rsd_decay`, `denit_exp` and `denit_frac` are also used by the carbon model.
- soils.sol: organic carbon, clay and silt of each layer.
- print.prt and carbon outputs: carbon output files, and the optional `carbon_layers.prt` that sets how many soil layers the per-layer carbon outputs contain.
- Soil carbon: description of the carbon model.

Sources

- Reader: carbon_bsn_read.f90, called from proc_bsn.f90
- Type definitions and module variables: `organic_fractions`, `carbon_water_coef`, `manure_coef`, `organic_controls` and the residue tunables in carbon_module.f90; tillage variables in tillage_data_module.f90; `photo_degrade_factor` in plant_data_module.f90
- File name slot: `in_basin%carbon_bsn` in input_file_module.f90, read in readcio_read.f90
- Pool initialisation: soil_nutcarb_init.f90, called from main.f90.in
- Soluble carbon losses: nut_orgnc2.f90
- Organic amendments: pl_fert.f90, pl_manure.f90
- Tillage and mixing: mgt_tillfactor.f90, mgt_biomix.f90, mgt_newtillmix_cswat1.f90, tillage day counter in hru_control.f90
- Decomposition factors: cbn_zhang2.f90, fcgd.f90
- Residue decomposition: cbn_surfrsd_decomp.f90
- `n_act_frac`: nut_nminrl.f90
- Calibration: cal_parm_select.f90

Column	Fortran variable
init_seq	org_frac%frac_seq
init_microb	org_frac%frac_hum_microb
init_slow	org_frac%frac_hum_slow
init_passive	org_frac%frac_hum_passive
koc_c	cb_wtr_coef%prmt_21
solc_ratio	cb_wtr_coef%prmt_44
till_eff_days	till_eff_days (integer)
manure_c_frac	man_coef%rtof
bio_consol	bio_consf
till_consol	till_consf
tmpf_eqn	org_con%tmpf (integer)
watf_eqn	org_con%watf (integer)
t_cbn_min, t_cbn_opt, t_cbn_max	org_con%tn, org_con%top, org_con%tx
bmix_a, bmix_b, bmix_c	bmix_a, bmix_b, bmix_c
tillmix_a, tillmix_b, tillmix_c	tillmix_a, tillmix_b, tillmix_c
sfc_rsd_photodeg	photo_degrade_factor
n_act_frac, cnr_cap, cnr_ref, cpr_cap, cpr_ref	module variables of the same names
mathers_method	mathers_int, stored as org_frac%mathers_method = (mathers_int == 1)

Purpose

carbon_lyr.bsn contains the decomposition rates of the litter, microbial and humus pools of the dynamic soil carbon model (carbon = 2 in codes.bsn), and the fractions of decomposed carbon released as CO2. There is one row for each of two layer groups:

- layer group 1: the first soil layer of each HRU. SWAT+ adds a 10 mm surface layer when the first layer in soils.sol is deeper than 19.5 mm, so this is normally the 10 mm surface layer.
- layer group 2: all other soil layers.

The basin-wide parameters of the carbon model are in carbon.bsn.

Activation and file name

The file is read only when carbon = 2, right after carbon.bsn, and it is then required. It is not listed in file.cio. Its name is built from the carbon.bsn name on the basin line of file.cio by inserting _lyr before the extension, so the default carbon.bsn gives carbon_lyr.bsn, and mybasin.bsn gives mybasin_lyr.bsn.

SWAT+ writes an error to the screen and to diagnostics.out and stops if the file does not exist, cannot be opened, or has fewer than two valid data rows.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped and may be blank. Each later line is one layer group: an integer layer group number followed by 11 values, in the order below. The reader matches values by position only.

#	Column	Units	Group 1 (example)	Group 2 (example)	Description
1	`layer`	-	1	2	Layer group: `1` first soil layer, `2` all deeper layers.
2	`hp_rate`	1/day	1.2e-05	1.2e-05	Decomposition rate of passive humus under optimal conditions.
3	`hs_rate`	1/day	2.92e-04	1.81e-04	Decomposition rate of slow humus under optimal conditions.
4	`microb_rate`	1/day	0.0164	0.02	Decomposition rate of microbial biomass under optimal conditions. Used only in group 2.
5	`meta_rate`	1/day	0.0405	0.0507	Decomposition rate of metabolic litter under optimal conditions.
6	`str_rate`	1/day	0.0107	0.0134	Decomposition rate of structural litter under optimal conditions, before the lignin reduction.
7	`microb_top_rate`	1/day	0.0164	0.02	Decomposition rate of microbial biomass in the first soil layer. Used only in group 1.
8	`hs_hp`	fraction	0.05	0.05	Fraction of decomposed slow humus that goes to passive humus, before the clay adjustment.
9	`a1co2`	fraction	0.60	0.55	Fraction of decomposed metabolic litter and non-lignin structural litter released as CO2.
10	`asco2`	fraction	0.55	0.55	Fraction of decomposed slow humus released as CO2.
11	`apco2`	fraction	0.55	0.55	Fraction of decomposed passive humus released as CO2.
12	`abco2`	fraction	0.55	0.0	Fraction of decomposed microbial biomass released as CO2. Used only in group 1.

The example values are those of `refdata/Ames_sub1/carbon_lyr.bsn` (see Example).

Rows are read until the end of the file. A row whose layer group number is below 1 or above 2 is skipped with a message in `diagnostics.out`. A row that cannot be read (a missing value, or text in a number column) ends the reading at that row without a message, and the rows after it are ignored. The rows may be in any order. The reader counts accepted rows without checking that both groups are present, so a file with two rows for group 1 and none for group 2 passes the check and leaves the group 2 values at 0, which stops decomposition in all layers below the first.

Use in the carbon model

Each day, for each soil layer, the potential transformation of each pool is its carbon times its rate from the layer's group times the combined temperature, water, oxygen and tillage factor (see `carbon.bsn`):

$$\Delta C_{pool} = k_{pool}\, f_{env}\, C_{pool}$$

with these additions:

- metabolic litter: `a1co2` is released as CO2 and the rest goes to microbial biomass.
- structural litter: the rate is reduced by the lignin fraction L of the litter (at most 0.8), $k = \text{str_rate} \cdot \exp(-3L)$. Of the decomposed lignin, 30% is released as CO2 and 70% goes to slow humus. Of the non-lignin part, `a1co2` is released as CO2 and the rest goes to microbial biomass.
- microbial biomass: the first soil layer uses `microb_top_rate`. Deeper layers use `microb_rate` times a texture factor $0.25 + 0.0075\,\text{sand}$ (sand in %).
- slow humus: the fraction going to passive humus is $\max(0.001,\ \text{hs_hp} - 0.00009\,\text{clay})$, with clay in %. `asco2` is released as CO2 and the rest goes to microbial biomass.
- passive humus: `apco2` is released as CO2 and the rest goes to microbial biomass.
- microbial biomass: in the first soil layer `abco2` is released as CO2. In deeper layers the CO2 fraction is calculated from sand content, $0.17 + 0.0068\,\text{sand}$, and the `abco2` of group 2 is not used. The fraction going to passive humus is $0.003 + 0.00032\,\text{clay}$ in all layers.

The actual transformations are reduced when there is not enough mineral N to meet the N demand of the receiving pools. See soil carbon for the full pool structure.

Calibration

The parameter names `hp_rate`, `hs_rate`, `microb_rate`, `meta_rate`, `str_rate`, `microb_top_rate`, `hs_hp`, `a1co2`, `asco2`, `apco2` and `abco2` are recognised in `calibration.cal` with object type `bsn`, and the layer group is meant to be given as the layer of the change. In 62.0.1, basin parameters are passed to the change routine without a layer number, so these changes are skipped and have no effect (see parameter change files). Calibrating these rates means editing `carbon_lyr.bsn`.

Example

`refdata/Ames_sub1/carbon_lyr.bsn` in the SWAT+ repository:

```
carbon_lyr.bsn: derived from carb_coefs.cbn (per-layer carbon coefficients)
           layer         hp_rate         hs_rate     microb_rate       meta_rate        str_rate microb_to
               1         1.2e-05        2.92e-04          0.0164          0.0405          0.0107
               2         1.2e-05        1.81e-04            0.02          0.0507          0.0134
```

The first soil layer decomposes metabolic and structural litter more slowly than the deeper layers (0.0405 and 0.0107 per day, against 0.0507 and 0.0134) and slow humus faster (2.92e-04 against 1.81e-04 per day). The 0.0 for `abco2` in group 2 has no effect, because deeper layers calculate that fraction from their sand content. `microb_rate` in group 1 and `microb_top_rate` in group 2 are not used.

Related files

- carbon.bsn: basin-wide carbon parameters; its name sets the name of this file.
- codes.bsn: `carbon = 2` turns the carbon model on.
- soils.sol: layer depths, clay and sand contents.
- Carbon outputs: per-layer pools, fluxes and CO2.
- Soil carbon: description of the carbon model.

Sources

- Reader: second part of carbon_bsn_read.f90
- Type definitions: `type carbon_inputs` (array `carbdb(2)`) and `type organic_allocations` (array `org_allo(2)`) in carbon_module.f90
- Use: cbn_zhang2.f90
- Surface layer: soils_init.f90
- Calibration: cal_parm_select.f90, cal_conditions.f90

Column	Fortran variable
layer	layer_id, index into carbdb and org_allo
hp_rate, hs_rate, microb_rate, meta_rate, str_rate, microb_top_rate, hs_hp	carbdb(layer)%... of the same names
a1co2, asco2, apco2, abco2	org_allo(layer)%... of the same names